
From Airspace to Markets

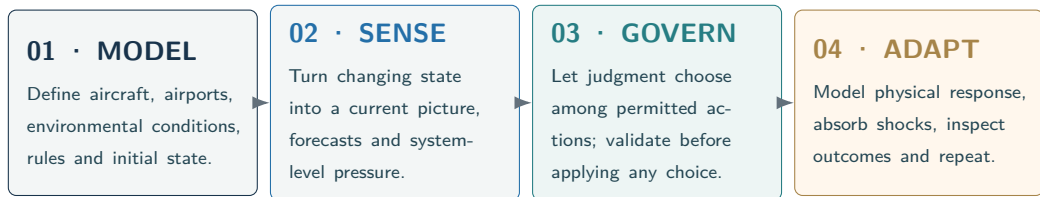
Governed Autonomous Control Architecture

A synthetic multi-airport simulator and its transfer to finance

Autonomy is not a model call. It is a governed loop of perception, prediction, judgment, execution and feedback.

One argument, told in four acts

Build a world, make its state legible, constrain action, then learn from what execution changes.



Airspace is the laboratory

Governance is the mechanism

Finance is the transfer test

The thesis: reasoning proposes; the system decides

The LLM contributes bounded judgment; deterministic rules and the world model retain authority over admissibility and outcome.



EVIDENCE

State estimates and forecasts help explain what is happening and what may happen next.

JUDGMENT

The model interprets a structured packet and recommends from a narrow menu.

AUTHORITY

Rules accept or reject proposals; dynamics, not prose, determine what occurs.

Build a reproducible control substrate

MODEL · 01

Layer 1 · Cells 2–4

A controller can only be evaluated if its starting world and software assumptions are explicit and repeatable.

INSIDE THIS LAYER

- ▶ Establish the runtime and required libraries.
- ▶ Define configuration, parameters and state containers.
- ▶ Initialize from documented seeds and reset procedures.

EVIDENCE PASSED FORWARD

A known initial state, configuration record and version information allow runs to be compared or replayed.

WHY IT MATTERS

Without a stable substrate, outcome differences may come from setup drift rather than control quality.

CONTROL BOUNDARY

Initialization defines the experiment; it does not grant decision authority to the model.

Separate the controller into accountable layers

The system is decomposed so that each component has a clear job, evidence contract and authority limit.

INSIDE THIS LAYER

- ▶ The world model owns state and physical transitions.
- ▶ Perception, prediction and analysis construct evidence.
- ▶ The LLM proposes; the harness validates; execution applies.

EVIDENCE PASSED FORWARD

A traceable chain from observed state to accepted action and resulting transition.

WHY IT MATTERS

Separation makes failures localizable: bad input, weak forecast, unsafe proposal, invalid rule or unexpected dynamics.

CONTROL BOUNDARY

No component may skip its downstream gate; the LLM is a supervised participant, never the top-level authority.

Create a world with interacting constraints

MODEL · 03

Layer 2 · Cells 6–10

The controlled object is a population sharing infrastructure, not an isolated aircraft moving in empty space.

INSIDE THIS LAYER

- ▶ Represent airports, aircraft, flight phases and routes.
- ▶ Track performance-relevant attributes and current targets.
- ▶ Model weather, wind, runway modes and available capacity.

EVIDENCE PASSED FORWARD

The simulator state combines agents, shared resources and environmental conditions at each time step.

WHY IT MATTERS

Interactions create conflicts and queues: one aircraft's action can affect other traffic and airport flow.

CONTROL BOUNDARY

Environmental conditions evolve independently; controller commands must respect modeled capabilities and constraints.

Perceive the world as structured state

SENSE · 01

Layer 3 · Cell 11

Perception turns dispersed simulation variables into one current, consistent operational picture.

INSIDE THIS LAYER

- ▶ Read each aircraft's identity, position, altitude, speed and phase.
- ▶ Normalize units, missing values and categorical states.
- ▶ Assemble a timestamped table for later analysis.

EVIDENCE PASSED FORWARD

A canonical snapshot: what objects exist, where they are, and which relevant conditions apply now.

WHY IT MATTERS

A clean state representation reduces ambiguity and makes downstream decisions comparable across runs.

CONTROL BOUNDARY

Perception should report observed or derived state, not quietly embed recommendations or authorize actions.

Predict trajectories, not just current positions

SENSE · 02

Layer 4 · Cell 12

A forecast turns a snapshot into a time-bounded estimate of emerging conflicts and capacity pressure.

INSIDE THIS LAYER

- ▶ Project current motion over a defined horizon.
- ▶ Compare projected paths and arrival windows.
- ▶ Flag proximity, sequencing pressure and likely constraint breaches.

EVIDENCE PASSED FORWARD

Forecast tracks, conflict candidates and timing estimates that can be inspected alongside current state.

WHY IT MATTERS

Earlier warning creates more room for gentle interventions and coordinated sequencing.

CONTROL BOUNDARY

A forecast is not a fact or a command. Preserve its horizon and uncertainty; re-predict after state changes.

Analyze the whole flow, not one aircraft

SENSE · 03

Layer 5 · Cell 13

System analysis aggregates local state into congestion, capacity use and sequencing pressure.

INSIDE THIS LAYER

- ▶ Summarize arrivals, departures and queues by airport or area.
- ▶ Compare demand with runway or sector capacity.
- ▶ Identify bottlenecks and interacting priorities.

EVIDENCE PASSED FORWARD

Aggregate flow indicators and ordered traffic lists that explain where pressure is building.

WHY IT MATTERS

A locally safe action can still worsen a queue elsewhere; system-wide measures expose these trade-offs.

CONTROL BOUNDARY

Priority criteria and capacity assumptions should be explicit, reviewable and separate from model-generated narrative.

Define the legal action space first

Governance is executable when each proposed action is checked against a fixed vocabulary and deterministic limits.

INSIDE THIS LAYER

- ▶ Enumerate permitted action types and aircraft identifiers.
- ▶ Define numeric ranges and state-dependent constraints.
- ▶ Check schema, eligibility, conflicts and invariants before application.

EVIDENCE PASSED FORWARD

A validation result with accepted fields or a rejection reason suitable for logging.

WHY IT MATTERS

A narrow action space reduces ambiguity, makes tests concrete and supports consistent enforcement.

CONTROL BOUNDARY

Unsafe, stale or malformed proposals fail closed; the model cannot change the rules.

Build decision context from evidence

The context packet should make the decision understandable without making the model responsible for data discovery.

INSIDE THIS LAYER

- ▶ Include the current state and relevant predicted events.
- ▶ State weather, capacity, policy constraints and uncertainty.
- ▶ Present only currently valid candidate actions in a stable schema.

EVIDENCE PASSED FORWARD

A compact, timestamped packet linking evidence, constraints and options for one decision cycle.

WHY IT MATTERS

Relevant context improves answer quality and limits reliance on unsupported assumptions.

CONTROL BOUNDARY

Context informs judgment but does not confer permission; candidate actions and limits come from the harness.

Use bounded judgment with a fallback

The model contributes synthesis where the situation is nuanced, while remaining replaceable and constrained.

INSIDE THIS LAYER

- ▶ Ask for a choice from the supplied candidates.
- ▶ Require a concise rationale tied to supplied evidence.
- ▶ Handle invalid output, timeout or unavailability with a known fallback.

EVIDENCE PASSED FORWARD

A proposed action, rationale and status that can be checked independently before use.

WHY IT MATTERS

Judgment can reconcile competing pressures and explain trade-offs that isolated rules may not capture.

CONTROL BOUNDARY

The model does not invent an action outside the menu. Failure reduces sophistication, not baseline safety.

Execute only what passed validation

ACT · 01

Layer 9 · Cell 19

Execution is a controlled state update, not an automatic continuation of generated text.

INSIDE THIS LAYER

- ▶ Match each accepted action to a known aircraft and permitted field.
- ▶ Apply target speed, heading, altitude or hold as specified.
- ▶ Record the applied command and validation result.

EVIDENCE PASSED FORWARD

The accepted intervention, updated targets and a linked action record.

WHY IT MATTERS

A visible proposal-to-execution boundary makes control auditable and prevents accidental command leakage.

CONTROL BOUNDARY

Rejected proposals never reach the state updater; accepted targets remain subject to physical dynamics.

Let dynamics determine the realized outcome

ACT · 02

Layer 10 · Cell 20

Aircraft respond over time; a target instruction is not an instantaneous state transition.

INSIDE THIS LAYER

- ▶ Apply acceleration, turn-rate and climb/descent limits.
- ▶ Advance position and state over the simulation step.
- ▶ Reconcile the resulting state with routes and handoff conditions.

EVIDENCE PASSED FORWARD

New positions, speeds and altitudes after modeled inertia and operational constraints.

WHY IT MATTERS

Only a realistic response model can show whether an accepted intervention actually helps.

CONTROL BOUNDARY

The dynamics layer owns feasibility and realized motion; commanded targets do not override physical limits.

Test against shocks the controller does not choose

ADAPT · 01

Layer 11 · Cell 21

Exogenous events challenge the policy by changing conditions independently of its decisions.

INSIDE THIS LAYER

- ▶ Schedule or inject weather and capacity changes.
- ▶ Update affected runway, route or operating constraints.
- ▶ Trigger fresh sensing, prediction and planning after the event.

EVIDENCE PASSED FORWARD

An event timeline, altered environmental state and the controller's response to each change.

WHY IT MATTERS

Favorable static conditions cannot establish robustness; changing conditions reveal recovery behavior.

CONTROL BOUNDARY

Shocks enter through the world model and are logged separately from controller actions.

Close the loop with fresh state

ADAPT · 02

Layer 12 · Cell 22

An autonomous loop exists only when action changes the world and the changed world is measured again.

INSIDE THIS LAYER

- ▶ Apply validated commands and advance the simulator.
- ▶ Rebuild the state snapshot and all dependent forecasts.
- ▶ Re-evaluate conflicts, queues and constraints before the next decision.

EVIDENCE PASSED FORWARD

A sequence of timestamped states, actions and post-action outcomes.

WHY IT MATTERS

Repeated feedback detects side effects and stale assumptions; it is the unit of closed-loop control.

CONTROL BOUNDARY

Never make a new decision from an old packet after execution or an environmental event.

Make decisions observable and contestable

ASSURE · 01

Layer 13 · Cell 23

Operators need evidence of both external conditions and the internal path from evidence to action.

INSIDE THIS LAYER

- ▶ Show aircraft, weather, capacity and conflict status.
- ▶ Capture proposals, validator outcomes, fallbacks and execution.
- ▶ Preserve timestamps, reasons and relevant model/system status.

EVIDENCE PASSED FORWARD

A human-readable view and machine-readable event history for inspection and replay.

WHY IT MATTERS

Observability supports debugging, safety review, accountability and comparison across scenarios.

CONTROL BOUNDARY

Do not log only successful actions; rejected proposals and fallback use are essential evidence.

Evaluate the integrated experiment

ASSURE · 02

Layer 14 · Cell 24

The full run tests whether the layers cooperate coherently through time, including under disturbance.

INSIDE THIS LAYER

- ▶ Reset to a recorded initial condition and execute each simulation step.
- ▶ Process shocks, forecasts, proposals, validation and dynamics in order.
- ▶ Aggregate safety, flow, fallback and audit measures at the end.

EVIDENCE PASSED FORWARD

Telemetry, state histories and outcome measures linked to a reproducible run configuration.

WHY IT MATTERS

Layer-level correctness is not enough; integration can expose timing, state-sync and handoff failures.

CONTROL BOUNDARY

Evaluate safety and resilience alongside throughput. A clean model response alone is not evidence of safe control.

Transfer to finance: preserve the architecture, change the objects

The financial analogue is not to let an LLM trade; it is to add bounded supervisory judgment to an established market-control stack.

Control function	Airspace model	Financial analogue
World state	Aircraft, airport and runway state	Positions, orders, cash, margin and venues
External conditions	Weather, wind and capacity changes	Volatility, liquidity, news and venue outages
Perception and forecast	State table, trajectories and conflicts	Exposure, slippage, liquidity gaps and drawdown
Candidate actions	Change target, sequence or hold	Hedge, resize, reroute, pause or allocate
Authority gate	Action menu and deterministic safety checks	Risk, compliance, mandate and order-size limits
Execution and dynamics	Target updates, inertia and handoff	Routing, fills, impact, P&L and exposure evolution
Audit evidence	State history, proposals and validation	Blotter, approvals, exceptions and surveillance

FAST, DETERMINISTIC CONTROL

Data normalization, exposure calculation, order admissibility, execution limits and kill switches belong in dependable systems.

SLOWER, BOUNDED JUDGMENT

Regime interpretation, exception triage and scenario synthesis can support human oversight without taking order authority.

Capability may expand. Permission must remain explicit.

The institutional design rule

Useful autonomy separates reasoning, authority, execution and feedback—and reconnects them with durable evidence.

1. Reason broadly Interpret context, exceptions and competing objectives.

2. Authorize narrowly Enforce candidate menus, limits, validation and fallback.

3. Execute realistically Respect physical or market dynamics and operational friction.

4. Learn visibly Re-perceive, monitor, preserve evidence and stress test.

Questions for deployment

- ▶ Is the state understandable and current?
- ▶ Can unsafe actions be rejected deterministically?
- ▶ Does the system degrade safely without the model?
- ▶ Can an auditor replay the decision and outcome?

Sources: Åström & Murray; Endsley; Leveson; NIST AI RMF.

LLM judgment stays inside—not above—the architecture.